# Exercise Sheet 11

## Integration — Network Science Across All Topics

5942UE Network Science

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# Module A — From Structure to Process: A Case Study

# Exercise 11.A.1: Network Characterization

Karate club graph — full structural profile.

1. Is the graph connected? What is the diameter?
2. Compute global and average clustering.
3. Top-3 nodes by degree, betweenness, closeness centrality. Same nodes in all three?
4. Are there any bridges?

**Solution:**

```
import networkx as nx

G = nx.karate_club_graph()

print(nx.is_connected(G), nx.diameter(G))
print(nx.transitivity(G), nx.average_clustering(G))

deg = nx.degree_centrality(G)
bet = nx.betweenness_centrality(G)
clo = nx.closeness_centrality(G)

print(sorted(deg, key=deg.get, reverse=True)[:3])
print(sorted(bet, key=bet.get, reverse=True)[:3])
print(sorted(clo, key=clo.get, reverse=True)[:3])

print(list(nx.bridges(G)))
```

The network is connected with diameter 5 and average clustering $\approx 0.57$. Nodes 0 and 33 (the two faction leaders) consistently rank highest across all centrality measures — both structural hubs and information brokers. The few bridges in the graph hint at the impending faction split.

# Exercise 11.A.2: Community and Social Structure

1. Detect communities via greedy modularity. How many? Modularity score?
2. Compare to the known `"club"` faction split. Does it match?
3. Compute the E-I index for the faction.
4. Treat within-community ties as positive, cross as negative — is the network structurally balanced?

**Solution:**

```
from networkx.algorithms.community import greedy_modularity_communities
import networkx as nx

G = nx.karate_club_graph()
comms = list(greedy_modularity_communities(G))
mod = nx.community.modularity(G, comms)
```

Greedy modularity finds $\approx 4$ communities with modularity $\approx 0.38$, broadly aligning with the faction split. The E-I index $\approx -0.5$ confirms strong intra-club preference. The faction structure is largely balanced — within-club triangles are all-positive, cross-club triangles fit "enemy of my enemy" patterns.

# Exercise 11.A.3: Ties, Paths, and Information Flow

1. Classify each edge as a strong tie ($\geq 2$ common neighbours) or weak tie. What fraction is weak?
2. Are any weak ties also bridges? What does this confirm about Granovetter?
3. Compare $L$ and $C$ to an Erdős–Rényi graph with the same $n$ and $m$. Is karate club a small-world network?

**Solution:**

```
import networkx as nx

G = nx.karate_club_graph()
def common_neighbors(u, v):
    return len(set(G.neighbors(u)) & set(G.neighbors(v)))

weak = [(u, v) for u, v in G.edges() if common_neighbors(u, v) < 2]
bridges = set(nx.bridges(G))
weak_bridges = [e for e in weak if tuple(sorted(e)) in {tuple(sorted(b)) for b in bridges}]
```

Bridges in the karate club are consistently weak ties — confirming Granovetter's hypothesis that local bridges are rarely reinforced by mutual neighbours. The graph has $C \gg C_{rand}$ but $L \approx L_{rand}$ — a classic small-world network combining tight local structure with rapid global reach.

# Module B — Advanced Dynamics and Robustness

# Exercise 11.B.1: Vaccination Strategy

You can vaccinate (immunize) only 5 of 34 nodes. Compare three strategies:

1. **Random:** 5 random nodes
2. **Hub:** top-5 by degree
3. **Bridge:** top-5 by betweenness

For each: run SIR ($\beta = 0.4$, seed = 2nd-highest-degree node, 50 runs). Average fraction infected?

When does the bridge strategy beat the hub strategy?

**Solution:**

```
import networkx as nx
import random

def sir_vaccinated(G, seed, vaccinated, beta=0.4, steps=20):
    state = {n: 'S' for n in G.nodes()}
    for v in vaccinated:
        state[v] = 'R'
    state[seed] = 'I'
    for _ in range(steps):
        ns = state.copy()
        for n in G.nodes():
            if state[n] == 'I':
                ns[n] = 'R'
            elif state[n] == 'S':
                if any(state[nb] == 'I' and random.random() < beta
                       for nb in G.neighbors(n)):
                    ns[n] = 'I'
        state = ns
    return sum(1 for s in state.values() if s == 'R') / G.number_of_nodes()

G = nx.karate_club_graph()
```

```python
hub_vacc    = [n for n, _ in sorted(G.degree, key=lambda x: x[1], reverse=True)[:5]]
bridge_vacc = [n for n, _ in sorted(nx.betweenness_centrality(G).items(),
                                    key=lambda x: x[1], reverse=True)[:5]]
```

Hub vaccination usually wins inside dense communities ($\approx 30$ infected vs $\approx 70$ random). But high-betweenness "bridge" nodes are key for stopping spread *between* factions. In networks with sparse links between dense groups, bridge vaccination can outperform hub vaccination.

# Exercise 11.B.2: Cascade vs. Epidemic

Compare on the karate club:

- **Simple contagion:** SIR, $\beta = 0.4$
- **Complex contagion:** threshold cascade, $\theta = 0.35$

For each, seed from node 0 (hub).

1. Which spreads faster to 50% adoption?
2. Which reaches more nodes total?
3. Now seed from node 11 (peripheral, degree 1). Does the comparison change?

**Solution:**

Hubs facilitate simple contagion almost instantly — SIR from a hub reaches most of the network within a few steps. But for complex contagion, a single hub seed is often *insufficient*: each neighbour sees only one adopter, below threshold.

Simple contagion is highly vulnerable to hub seeding anywhere; complex contagion is much more resilient and requires clustered "social proof". Seeding from a peripheral node makes simple contagion noticeably slower but barely changes complex contagion — what matters for cascades is local *density*, not the seed's centrality.

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# Exercise 11.B.3: Full Network Pipeline (Capstone)

Build a complete pipeline that, for any input graph, reports basic stats, clustering, top centrality, communities, modularity, and a quick SIR epidemic from the hub. Run it on:

1. Karate club
2. `nx.barabasi_albert_graph(50, 2)`
3. `nx.watts_strogatz_graph(50, 4, 0.1)`

(a) Which graph is most clustered? Smallest diameter? (b) Which has highest epidemic vulnerability from the hub? (c) Which best models real social networks?

**Solution:**

```
import networkx as nx
import random
from networkx.algorithms.community import greedy_modularity_communities

def analyze_network(G, name="Network"):
    print(f"\n=== {name} ===")
    print(f"n={G.number_of_nodes()} m={G.number_of_edges()}")
    if nx.is_connected(G):
        print(f"D={nx.diameter(G)} L={nx.average_shortest_path_length(G):.3f}")
    print(f"C_global={nx.transitivity(G):.3f}  C_avg={nx.average_clustering(G):.3f}")
    deg = nx.degree_centrality(G)
    hub = max(deg, key=deg.get)
    comms = list(greedy_modularity_communities(G))
    print(f"hub={hub}  communities={len(comms)}  Q={nx.community.modularity(G, comms):.3f}")

    def quick_sir(G, seed, beta=0.3, steps=15):
        state = {n: 'S' for n in G.nodes()}
        state[seed] = 'I'
        for _ in range(steps):
            ns = state.copy()
            for n in G.nodes():
```

```python
            if state[n] == 'I':
                ns[n] = 'R'
            elif state[n] == 'S':
                if any(state[nb] == 'I' and random.random() < beta
                       for nb in G.neighbors(n)):
                    ns[n] = 'I'
```

Barabási–Albert has small diameter via hubs but low clustering. Watts–Strogatz has high clustering and slightly larger paths. Real networks like the karate club combine both — hub-driven small diameter *and* dense local communities. This "clustered small world" is the most realistic model for human social structure.

# Exercise 11.B.4: Concept Map — Synthesis

No code. Connect concepts across the course:

1. **L1–L2:** Why does connectivity matter for epidemics?
2. **L3–L8:** Granovetter's weak ties are bridges. How do they affect simple vs. complex contagion differently?
3. **L4–L8:** When do high-betweenness nodes matter for vaccination, and when do they not?
4. **L5–L8:** How does homophily affect complex contagion?
5. **L7–L8:** Which small-world property helps simple contagion? Which helps complex contagion?

**Solution:**

**1. Connectivity & epidemics:** A disconnected graph confines epidemics; within a component, diameter bounds full spread time, and giant-component size bounds the population at risk.

**2. Weak ties:** Bridges *amplify* simple contagion (long-range shortcuts). They *block* complex contagion — a node on the far side of a bridge sees only one adopter and can't meet its threshold. Complex cascades stay within dense clusters.

**3. Betweenness:** High-betweenness nodes are critical for inter-community choke-points. But within-community hubs may have low betweenness despite high degree. Bridge vaccination stops between-group spread; hub vaccination stops within-group explosions.

**4. Homophily & complex cascades:** Homophily creates dense within-group edges → cascades sweep clusters quickly (threshold easily met) but stall at sparse cross-group ties. Complex cascades remain **confined to homophilous clusters**

unless seeded in multiple groups.

**5. Small worlds: Short paths** help **simple contagion** (few hops to reach everyone). **High clustering** helps **complex contagion** (dense triangles provide reinforcing adopters). Neither alone explains spreading; structure and process rules interact.

# Wrap-Up

- structure and dynamics interact: same graph, different process rules → different outcomes
- centrality is not a single concept; the right measure depends on the question
- targeted interventions (vaccination, seeding) require matching the strategy to the contagion type
- "small world + clustered communities" is the realistic baseline for most human networks